Shivam Tiwari

+91-7905032556 shivamneo33@gmail.com linkedin.com/in/shivamtiwari3 github.com/shivamt21297 Google Scholar

Summary

Senior Software Engineer with **6+ years** of experience architecting and shipping mobile applications at scale (**180M+ MAU**). Deep expertise in **Native Android (Kotlin/Java)** and **React Native**, with additional hands-on experience across **iOS, backend, web**, and AI-powered development — spanning performance engineering, scalable system design, and full-stack delivery. Experienced technical leader — owning architecture decisions, mentoring engineers, and driving CI/CD — across high-growth product companies. Background in ML/Deep Learning research; actively applies **AI-assisted development workflows** to accelerate engineering velocity.

Metrics Snapshot

- **53% faster builds**
  30 min → 14 min
- **Cross-functional tech leadership**
  Android, RN, iOS, backend, CI/CD
- **15MB+ APK size reduction**
  Multiple modules DFM (ShareChat)
- **CriQ built from scratch**
  Full-stack, 0 → 1
- **App-wide framework owner**
  Intervention system (ShareChat)
- **AI workflows and On-device ML**
  Built PR review, test-gen AI agents and integrated ML models on-device.

Experience

Dream11 — Mumbai, India
Senior Software Engineer — *Dec 2024 – Present*

- **Led the 0→1 development of CriQ**, owning **React Native**(owning both **Android** and **iOS**), backend services, CI/CD, and release pipelines end-to-end — supporting separate builds for Indian and international markets with social sign-in.
- **Reduced CI and local build time by 53%** (30 min → 14 min) via Gradle optimizations, dependency caching, and modularization; established **GitHub Actions** pipelines for automated releases and **CodePush** OTA updates.
- **Integrated ML models on-device**: Compact models running real-time inference on device for **contest recommendations**, etc.
- **Built AI engineering workflows in production**: agentic automated PR reviews and unit-test generation.
- Integrated **Speech-to-Text and Text-to-Speech** in both CriQ and the Dream11 flagship app, enabling conversational and accessibility features.
- **Mentored junior and mid-level engineers** through structured code reviews, technical 1:1s, and pairing sessions, raising team-wide engineering standards.
- Led **architecture and design reviews**, driving best practices for scalability, modularity, and long-term codebase health.
- Drove **technical roadmap and architecture discussions** across Android, React Native, backend, and platform teams — influencing engineering direction across multiple cross-functional initiatives.

ShareChat & Moj — Bangalore, India
Senior Software Engineer – Android — *Oct 2021 – Dec 2024*

- **Architect of the Intervention Framework** — a lifecycle-aware, app-wide system managing all overlays (popups, tooltips, snackbars); used **Kotlin Coroutines Actor**, **MVI**, and **Dagger-Hilt** for queue orchestration and remote-controlled display logic; adopted company-wide for UX consistency.
- **Drove sustained crash-free rate improvements** across chat and chatroom modules through systematic ANR triage, memory profiling, and targeted fixes — contributing to platform stability.
- Built and shipped **monetization features for audio/video livestreaming** using **Firebase Firestore** and **Agora RTC**, directly impacting revenue.
- Migrated high-footprint features and modules like FFMPEG and Agora to **Dynamic Feature Modules (DFMs)**, reducing app install size and improving first-install conversion rate.
- Delivered production UIs in **Jetpack Compose** and **XML** across **MVVM / MVI / MVP**; wrote UI automation suites in **Maestro** and unit tests with **JUnit / Mockk**.

Toppr / BYJU's Hyderabad, India
Software Engineer – Android *Aug 2020 – Oct 2021*

- **Delivered BYJU's Future School from zero to launch in one month** with a team of 10 — animated educational videos, interactive questions, and gamified learning on an aggressive timeline.
- Co-developed **Toppr School OS** (team of 4), a full virtual schooling platform; owned Android architecture using **MVVM**, **Dagger-Hilt**, and XML. **Led the Android sub-team** alongside development responsibilities.

Technical Skills

Android: Kotlin, Java, Jetpack Compose, XML, MVVM / MVI / MVP, Coroutines & Flow, Dagger-Hilt, Room, Dynamic Feature Modules, WorkManager, Paging3

Cross-Platform: iOS, Swift, React Native, Expo, CodePush, React.js, TypeScript

Testing: JUnit, Mockito, Mockk, Maestro (UI Automation)

Backend & Web: Node.js, SQL, RESTful APIs, Proto3/gRPC

AI / ML: On-device ML, Agentic Workflows for Automated Code Review and Unit tests, Deep Learning, Speech-to-Text, Cursor, Claude Code

Infrastructure & Tools: Git, GitHub Actions, Firebase, GCP, AWS, Agora RTC, Datadog, JIRA, Confluence

Education

Indian Institute of Information Technology (IIIT) Guwahati Guwahati, India
B.Tech in Computer Science – CGPA: **8.68 / 10** *July 2016 – Dec 2020*

- Programming Club Head — organised intra/inter-institute coding competitions, ran problem-solving sessions.
- ACM-ICPC Regional 2018 — rank 61/128 teams. 4⋆ CodeChef rating. Runner-up, NITS-HACKS 2018.

Internships & Early Career

Indian Statistical Institute (ISI) Kolkata Kolkata, India
Research Intern – Deep Learning *May 2018 – Jul 2018*

- Researched unsupervised representation learning using **Autoencoders**; explored latent space structure for feature extraction on sensor and image data under faculty supervision at one of India's premier research institutions.

SJTech Solutions Pvt. Ltd. Bhopal, India
Part-time Data Scientist *May 2019 – Jan 2020*

- Built and deployed computer vision models for **object detection & localisation** (garbage, hand-pump detection in web-scraped imagery), **face/gender recognition**, and **real-time drowsiness & motion detection** from live camera feeds using Python and deep learning frameworks.

Research Publications (Full list on Google Scholar)

Enhanced Annotation Framework for Activity Recognition via Change Point Detection *IIIT Guwahati*

- Designed an S-CPD (Segmented Change Point Detection) framework to learn activity change points from unannotated sensor datasets and generate labels automatically.

Semi-supervised Subject Recognition in Low-Modal Sensor Data *IIIT Guwahati*

- Built two semi-supervised frameworks for subject recognition across ubiquitous and vision-sensor datasets under low-modality, scarce-label conditions.

Behavior Analysis via Routine Cluster Discovery in Ubiquitous Sensor Data *IIIT Guwahati*

- Modelled latent feature distributions for user-specific behavioural characterisation from ubiquitous sensor data.

Deep Unsupervised Methods for Behavior Analysis in Ubiquitous Sensor Data *IIIT Guwahati*

- Developed a self-supervised approach for latent behavioural feature learning; demonstrated improved user modelling without labelled data.

Hyperspectral Image Classification using Semi-supervised Deep Learning

- Built a deep learning pipeline using clustering-generated pseudo-labels for hyperspectral image classification, reducing dependence on manual ground-truth annotation.